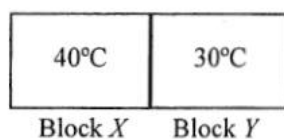


Book 1 section 2.3: DSE multiple choice

2012 Q1

1. Two metal blocks X and Y of the same mass and of initial temperatures 40°C and 30°C respectively are in good thermal contact as shown. The specific heat capacity of X is greater than that of Y . Which statement is correct when a steady state is reached? Assume no heat loss to the surroundings.



- A. The temperature of block X is higher than that of block Y .
- B. Their temperature becomes the same and is lower than 35°C .
- C. Their temperature becomes the same and is higher than 35°C .
- D. Their temperature becomes the same and is equal to 35°C .

PP Q3

3. Peter adds 50 g of milk at 20°C to 350 g of tea at 80°C , what is the final temperature of the mixture?

Given: Specific heat capacity of milk = $3800 \text{ J kg}^{-1} \text{ }^{\circ}\text{C}^{-1}$
Specific heat capacity of tea = $4200 \text{ J kg}^{-1} \text{ }^{\circ}\text{C}^{-1}$

- A. 50.0°C
- B. 72.5°C
- C. 73.1°C
- D. 77.4°C

2017 Q1

1. 30 g of milk at 10°C is added to 120 g of coffee at 80°C . Assuming there is no heat loss to the surroundings, what is the final temperature of the mixture?

Given: specific heat capacity of milk = $3800 \text{ J kg}^{-1} \text{ }^{\circ}\text{C}^{-1}$
specific heat capacity of coffee = $4200 \text{ J kg}^{-1} \text{ }^{\circ}\text{C}^{-1}$

- A. 64.8°C
- B. 65.2°C
- C. 66.0°C
- D. 67.1°C

2018 Q3

3. Milk at 5°C is added to a cup of tea at 25°C . Which statements below are correct? Neglect the heat capacity of the cup and assume that there is no heat exchange with the surroundings.

- (1) The average kinetic energy of the water molecules in the tea decreases.
- (2) The average potential energy of the water molecules in the tea remains unchanged.
- (3) The energy lost by the tea equals the energy gained by the milk.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

Book 1 section 2.3: DSE multiple choice

2019 Q1

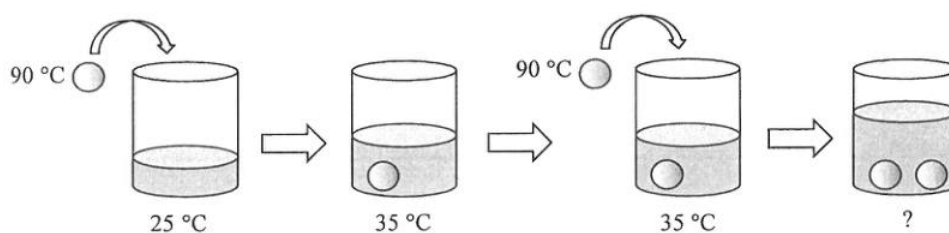
1. A block measuring 80°C is put into water at a temperature of 40°C . The final temperature of the mixture is 70°C . Which of the following deductions must be correct? Assume that there is no heat loss to the surroundings.
- A. The energy gained by the water is greater than the energy lost by the block.
 - B. The mass of the water is greater than the mass of the block.
 - C. The specific heat capacity of water is smaller than that of the block's material.
 - D. The heat capacity of the water is smaller than that of the block.

2022 Q1

1. A well-insulated vessel with negligible heat capacity contains warm water at 45°C . The temperature of warm water drops by 5°C after mixing with 50 g water at temperature 0°C added to the vessel. Find the original mass of warm water in the vessel.
- A. 400 g
 - B. 450 g
 - C. 500 g
 - D. 550 g

2026 Q3

3.



A metal sphere at 90°C is put into a cup of water. The temperature of the water increases from 25°C to 35°C . If a second identical metal sphere, also at 90°C , is added to the water, what is the final temperature of the water? Assume there is no heat loss to the cup or the surroundings throughout the process.

- A. 41.5°C
- B. 42.3°C
- C. 43.5°C
- D. 45.0°C